LINGUAFORCE: A Human-Annotated Benchmark for Pressure and Influence Strategies in Dialogue

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Abstract

Pressure in dialogue can take the form of a threat, an appeal to duty, or an attempt to make someone feel guilty. A binary label identifies its presence but does not describe how it is expressed. LINGUAFORCE provides human annotations for 4,066 English dialogues at four levels: pressure presence, overall intensity, fifteen non-exclusive influence strategies, and seven dimensions of expressed pressure. Ordinary requests and reasoned persuasion are included to distinguish influence from negative pressure. Twenty annotators completed the annotation batches. The dataset contains 3,144 dialogues labeled as exerting pressure and 922 labeled as not exerting pressure. Its labels support four prediction tasks. On 634 held-out dialogues, three zero-shot language models and two supervised lexical baselines obtain pressure AUROC between 0.947 and 0.958. Strategy macro-F1 remains between 0.570 and 0.610, and ordinal prediction separates the methods more clearly than detection. An additional matched Qwen3-4B fine-tuning diagnostic produces only small, task-specific changes, highlighting the difficulty of the fine-grained labels. The benchmark supports evaluation of both pressure recognition and its fine-grained description.

1 Introduction

Could you stay a little longer? and *If you leave, you clearly do not care about this family* ask for the same action. The second makes refusal carry a moral and emotional cost. Identifying this difference requires attention to the requested action, the surrounding exchange, and the consequences attached to refusal.

This distinction also concerns language as social action: an utterance can shape what a listener is expected or permitted to do, rather than merely describe a state of affairs (Ahearn, 2001).

Dialogue datasets study related phenomena such as mental manipulation, condescension, and toxicity (Wang et al., 2024; Khanna et al., 2025; Wang and Potts, 2019; Lin et al., 2023). Toxic language is not necessary for pressure, and a direct request is not necessarily coercive. A benchmark of conversational pressure therefore needs labels that describe both whether pressure is present and how it is expressed.

LINGUAFORCE pairs four kinds of human judgments. A binary label records whether a dialogue exerts negative pressure. An ordinal label records its overall intensity. Strategy labels describe the means used to influence the listener. Seven dimension scores describe directive force, option constraint, normative pressure, emotional pressure, deceptiveness, toxicity, and explicitness. A strategy identifies a kind of behavior, while a dimension records its degree or manner of expression.

The scope is pressure expressed in text. The annotations do not establish whether a listener actually complied or experienced harm. Ordinary requests, advice, and reasoned persuasion are included as controls; beneficial influence such as encouragement is not a separate target of this release.

The contributions are a human-annotated resource with four related label sets, operational definitions that distinguish pressure from ordinary influence, and an evaluation design covering detection, intensity, strategy recognition, and dimension prediction. Evaluation compares three zero-shot language models with two supervised lexical baselines and includes a matched before/after fine-tuning diagnostic. Strong detection scores coexist with lower strategy recognition scores, motivating evaluation beyond the presence of pressure alone.

2 Related Work

Influence strategies. Compliance-gaining research studies how speakers seek cooperation (Marwell and Schmitt, 1967; Wiseman and

Schenck-Hamlin, 1981). Reciprocity, authority, and social proof describe common sources of influence (Cialdini, 1984). Speech-act theory distinguishes requests, orders, and promises (Austin, 1962; Searle, 1969). Normative and moral appeals have also been studied as moral-foundation signals in language and text classification (Graham et al., 2009; Johnson and Goldwasser, 2018). These accounts motivate the strategy inventory. They do not imply that every request, promise, or appeal to authority is harmful.

Language datasets. MentalManip and MultiManip study manipulation in conversations (Wang et al., 2024; Khanna et al., 2025). Model-assisted augmentation and distillation have also been explored for mental-manipulation detection (Gao et al., 2025). SemEval-2023 Task 3 annotates persuasion techniques in online news (Piskorski et al., 2023). TalkDown and ToxicChat address condescension and toxicity (Wang and Potts, 2019; Lin et al., 2023), while work on microaggressions illustrates the difficulty of locating subtle interpersonal harm in text (Breitfeller et al., 2019). LINGUAFORCE focuses on dialogue-level pressure and pairs strategy categories with ordinal judgments. Its evaluation asks whether models can recover these human judgments from dialogue text.

Modeling choices. Contextual encoders such as BERT and RoBERTa are standard supervised text baselines (Devlin et al., 2019; Liu et al., 2019). We use TF–IDF models instead to make the lexical-signal comparison explicit and to keep the supervised baseline separate from the prompted language-model evaluation.

3 Annotation Scheme

3.1 Pressure and Intensity

Negative pressure is an attempt to obtain an action or decision by making refusal costly, unacceptable, or difficult to consider. Evidence includes threatened consequences, guilt, obligations used as leverage, misleading claims, and hostility directed toward obtaining compliance. Information exchange, ordinary advice, and disagreement do not automatically qualify. Annotators consider the complete dialogue rather than individual keywords. A listener's resistance does not make the pressure disappear.

The binary label records presence or absence. Overall intensity uses six levels: none (0), very slight (1), weak (2), moderate (3), strong (4), and very strong (5). Intensity concerns the degree of pressure, not the number of strategies. In the completed annotations, absence corresponds to intensity 0 and presence to intensity 1–5.

3.2 Influence Strategies

Table 1 lists fifteen non-exclusive labels. Multiple strategies may occur in one dialogue. Requests and rational persuasion are included because the task must distinguish ordinary influence from pressure. A strategy label alone does not determine the binary label.

Gaslighting is included under deception.[1] Deception requires contextual evidence of misleading facts, claims, or promises. An unverified assertion or an ordinary future promise is not automatically deceptive.

3.3 Seven Dimensions

Each dimension receives an ordinal score of 0 (absent), 1 (low), 2 (moderate), or 3 (high). Table 2 defines the seven judgments.

These scores are not interchangeable with overall intensity. A request can be explicit while leaving refusal available. An ultimatum can restrict options without containing insults. Normative and emotional pressure can co-occur in the same appeal. The scheme permits these overlaps and does not assume statistical independence between dimensions.

4 Dataset and Human Annotation

4.1 Composition

The dataset contains 4,066 English dialogues. The current partition has 3,432 development-pool dialogues and 634 test dialogues (Table 3). A fixed, intensity-stratified split divides the development pool into 3,088 training and 344 validation dialogues.

The dialogues were manually collected, organized, and written in English around common everyday interaction scenarios. They are not a corpus of verbatim real-world chat logs. This scenario-based collection provides examples of different influence strategies, rather than a representative sample of their frequency in everyday conversation. Dialogue construction and the annotation procedure below are separate stages.

[1] Gaslighting involves denial or distortion that leads someone to doubt their memory or judgment. A disagreement about memory alone is insufficient evidence.

Strategy	Annotation focus	Count
Request/Advice	An open request, suggestion, or recommendation.	1,644
Directive/Order	An instruction that expects compliance.	2,078
Rational Persuasion	Reasons or evidence offered to support an action.	506
Promise/Inducement	A benefit, reward, or favor offered in return for action.	258
Threat/Warning	An adverse consequence used to obtain action.	705
Moral Appeal/Guilt	An appeal to conscience or guilt.	1,851
Obligation/Duty	A role or responsibility invoked to obtain compliance.	1,712
Value Judgement/Shaming	A negative evaluation of the listener's character.	1,336
Authority	Rank, expertise, or status used to demand compliance.	340
Conformity/Social Proof	Others' behavior or agreement used as leverage.	893
Reciprocity/Debt	A past favor invoked to demand repayment through action.	737
Emotional Manipulation	Emotional vulnerability exploited to change a decision.	882
Deception/Gaslighting	Facts or memory distorted to influence the listener.	30
Logical Trap/False Dilemma	A misleading argument or false restriction of alternatives.	234
Verbal Abuse/Toxicity	Insults or hostility used to obtain compliance.	947

Table 1: Strategy definitions and positive counts across 4,066 dialogues. Labels may overlap, so counts do not sum to the number of dialogues.

Dimension	Judgment
Directive force	Strength of the push toward an action.
Option constraint	How much the listener's alternatives are narrowed.
Normative pressure	Leverage through duty, morality, or social norms.
Emotional pressure	Guilt, fear, shame, or pity used as leverage.
Deceptiveness	Misleading facts or promises supported by context.
Toxicity	Insults, hostility, or verbal abuse.
Explicitness	How directly the action or pressure is expressed.

Table 2: Human-rated dimensions, each scored from 0 to 3.

Partition	Total	Pressure	No pressure
Development pool	3,432	2,640	792
Test	634	504	130
All	4,066	3,144	922

Table 3: Counts from the completed human annotation sheets. The development pool is reserved for training and validation.

4.2 Annotation Procedure

Twenty computer science graduate students, all proficient in reading English dialogue, volunteered as annotators. All read the same annotation guidelines and completed their assigned batches independently without consulting other annotators' labels. Each dialogue received one annotation across twenty non-overlapping batches. Each sheet contained the dialogue and fields for the binary label, intensity, seven dimensions, fifteen strategies, and optional notes. The annotation package did not include model predictions. Its instructions asked annotators to read the full dialogue, assign presence and intensity, rate the dimensions, and select all applicable strategies.

Intensity	0	1	2	3	4	5
Count	922	232	511	1,460	775	166

Table 4: Distribution of human-rated overall intensity.

4.3 Quality Checks

The completed sheets were checked for coverage, valid ranges, repeated records, and consistency between binary presence and intensity. All 4,066 records matched uniquely to their source dialogues using identifier and text. All required cells are populated and in range. No binary–intensity contradictions were found. No exact duplicate dialogues were found after lowercasing and whitespace normalization. These checks detect formatting and consistency problems; they do not measure agreement between people. No dialogues have yet received independent duplicate annotations, so inter-annotator agreement cannot currently be estimated.

5 Dataset Characteristics

Pressure is present in 3,144 dialogues (77.3%). Table 4 shows the intensity distribution. Moderate pressure is the most frequent level. Majority-label baselines and metrics sensitive to class imbalance are therefore necessary for interpreting predictive performance.

Strategy frequency varies substantially (Ta-

ble 1). Directives and moral appeals are common, whereas contextual evidence for deception is rare. Micro-averaged performance alone would obscure errors on rare labels, so evaluation includes macro-F1 and per-label support. Dimension evaluation likewise reports each dimension rather than only an average. These frequencies describe this collection, not the prevalence of such behaviors in everyday conversation.

6 Benchmark Tasks and Evaluation

6.1 Prediction Tasks

Every task takes dialogue text as input. Human labels are evaluation targets and are not supplied as test-time features.

Pressure detection. Predict binary pressure. Report positive-class F1 and AUROC from a model's pressure score. Binary decisions are evaluated directly.

Intensity prediction. Predict a 0–5 level. Report mean absolute error (MAE) and quadratic-weighted κ (QWK). MAE measures error size, while QWK penalizes larger ordinal disagreements.

Strategy recognition. Predict fifteen non-exclusive labels. Report macro- and micro-F1, with per-strategy precision, recall, F1, and support in supplementary results.

Dimension prediction. Predict seven 0–3 ratings. Report MAE and QWK per dimension and their unweighted averages. Undefined QWK in degenerate evaluation subsets must be reported rather than replaced with an arbitrary value.

6.2 Baselines

The comparison includes GPT-6 Astra, DeepSeek-Flash, and Qwen3.8-Max, accessed through a company gateway, the DeepSeek API, and DashScope, respectively. Each receives the same label definitions, dialogue text, and output schema without labeled examples. The schema requests a binary decision and pressure score, intensity, strategy labels, and dimension ratings. Outputs are checked against the allowed ranges, with invalid outputs and retries reported.

Supervised baselines use word unigram/bigram or character 3–5-gram TF–IDF features, capped at 50,000 features with minimum document frequency 2 and sublinear term frequency. Logistic regression predicts pressure and fifteen independent strategy labels; ridge regression predicts intensity and each dimension. Ordinal regression outputs are clipped to the allowed range and rounded. The constant baseline ignores dialogue text, predicting training majorities for binary labels and training medians for ordinal labels. All fitted components use training data only.

Qwen and DeepSeek use temperature 0 with thinking disabled. GPT uses the gateway's default sampling settings and a fixed low reasoning budget. Because chain-of-thought prompting can change generated answers (Wei et al., 2022), the reported comparison fixes the output instruction and does not treat hidden reasoning as an evaluation target. These three API models receive a 1,600-token output budget; GPT's budget also includes reasoning tokens. Technical reports describe the broader OpenAI, Qwen, and DeepSeek model families (OpenAI, 2023; Yang et al., 2025; DeepSeek-AI et al., 2024); the names here are provider aliases, and those reports do not identify the exact deployed versions. Returned identifiers and usage are saved. These provider-specific configurations are not matched reasoning budgets.

Matched fine-tuning diagnostic. To test whether the annotations support supervised adaptation, we fine-tuned Qwen3-4B on the 3,088 training dialogues with LoRA. The run used rank 16, scaling factor 32, dropout 0.05, learning rate 10^{-4}, three epochs, batch size 1 with gradient accumulation 8, and a maximum sequence length of 2,048 tokens. The resulting checkpoint and the original Qwen3-4B received the same 634 test dialogues, output schema, and `/no_think` suffix. This matched comparison is reported as a diagnostic; it is separate from the zero-shot and lexical baseline comparison.

6.3 Experimental Controls

The split and lexical baselines use seed 2026. Validation AUROC selects pressure regularization from $C \in \{0.1, 1, 10\}$; validation F1 selects its decision threshold from 0.10 to 0.90 in increments of 0.01. Strategy macro-F1 selects C from the same grid and a shared threshold from $\{0.3, 0.4, 0.5, 0.6\}$. Each ordinal target selects ridge regularization from $\{0.1, 1, 10\}$ by validation MAE.

The three API baselines share an English user prompt and test dialogues. Invalid outputs are re-

Model	Valid n	AUROC	F1
GPT-6 Astra	634	0.947	0.959
Qwen3.8-Max	634	0.958	0.953
DeepSeek-Flash	634	0.954	0.950
Word TF–IDF	634	0.949	0.942
Character TF–IDF	634	0.948	0.943
Constant	634	0.500	0.886

Table 5: Pressure detection. GPT, Qwen, and DeepSeek are zero-shot; TF–IDF models use human training labels. Constant predicts the training-majority label. Valid counts are out of 634; AUROC additionally requires a valid pressure score (see released metrics).

Model	Valid n	MAE ↓	QWK ↑
GPT-6 Astra	634	0.610	0.791
Qwen3.8-Max	634	0.891	0.714
DeepSeek-Flash	634	0.997	0.630
Word TF–IDF	634	0.683	0.736
Character TF–IDF	634	0.642	0.765
Constant	634	1.114	0.000

Table 6: Overall intensity prediction on valid 0–5 outputs. Counts are out of 634. The constant predictor uses the training median.

Model	Valid n	Macro-F1	Micro-F1
GPT-6 Astra	634	0.610	0.685
Qwen3.8-Max	634	0.590	0.667
DeepSeek-Flash	634	0.595	0.677
Word TF–IDF	634	0.573	0.738
Character TF–IDF	634	0.570	0.732
Constant	634	0.046	0.231

Table 7: Fifteen-label strategy recognition. A task-valid response supplies all fifteen binary labels. One unambiguous key spelling is normalized without changing its value.

tained, with validity evaluated separately for each task. Out-of-range language-model ratings are not clipped or silently replaced. Metrics on valid outputs must be read together with their coverage; comparisons on differing subsets do not establish a method ranking. Syntax-only normalization closes one missing outer JSON brace in nine GPT answers and maps one unambiguous strategy-key spelling in 146 Qwen answers. These rules apply to all three API baselines and leave label values unchanged; raw responses remain available. Transient GPT transport errors are retried up to three times with increasing delays; invalid generated answers are not retried. Reproduction materials include split identifiers, prompts, settings, predictions, and scoring code. Lexical baseline results use one fixed seed. We estimate 95% confidence intervals using 1,000 paired dialogue-level bootstrap resamples (seed 2026). These quantify test-sample uncertainty for fixed predictions, not variation across training or API runs. Appendix A reports the full intervals and a diagnostic of false positives on non-pressuring requests and persuasion.

7 Results and Analysis

7.1 Pressure Detection

Table 5 reports detection on all 634 test dialogues. Qwen has the highest numerical AUROC (0.958), while GPT has the highest F1 (0.959). Word and character TF–IDF reach AUROC 0.949 and 0.948. For each language model, the paired 95% bootstrap interval for the AUROC difference from word TF–IDF includes zero. The data therefore support strong lexical predictability, without establishing a clear ranking advantage for prompted language models. GPT's F1 difference from word TF–IDF is 0.017, with a paired interval of [0.005, 0.031] under the reported decision rules.

Requests without pressure. Of the 130 no-pressure test dialogues, 126 carry human request/advice or rational-persuasion labels. Negative examples therefore include ordinary influence attempts, rather than predominantly casual conversation. The prompted and trained models misclassify 22.2%–34.1% of these dialogues (Appendix A, Table 11), showing that distinguishing pressure from ordinary influence remains challenging.

7.2 Intensity and Strategy Prediction

GPT achieves intensity MAE 0.610 and QWK 0.791, compared with 0.642 and 0.765 for character TF–IDF (Table 6). Their MAE difference is not conclusive (paired 95% CI: $[-0.104, 0.036]$). Qwen and DeepSeek have larger intensity errors despite similar pressure detection scores, indicating that detecting pressure and estimating its degree pose different challenges.

GPT leads strategy macro-F1 (0.610), while word TF–IDF leads micro-F1 (0.738; Table 7). Both differences between these models persist under paired resampling (Appendix A). The ranking depends on whether strategies receive equal weight or their decisions are pooled, motivating both metrics for this uneven inventory. Per-strategy scores and support counts are available in

Model	Valid n	MAE ↓	QWK ↑
GPT-6 Astra	634	0.430	0.606
Qwen3.8-Max	295	0.631	0.540
DeepSeek-Flash	634	0.636	0.547
Word TF–IDF	634	0.422	0.631
Character TF–IDF	634	0.424	0.614
Constant	634	0.760	0.000

Table 8: Dimension prediction on complete, range-valid seven-dimensional outputs. MAE and QWK are averaged over dimensions. Different coverage prevents a direct ranking; Table 9 uses identical dialogues.

Model	MAE ↓	QWK ↑	All exact ↑
GPT-6 Astra	0.393	0.595	0.667
Qwen3.8-Max	0.631	0.540	0.331
DeepSeek-Flash	0.593	0.524	0.506
Word TF–IDF	0.412	0.656	0.618
Character TF–IDF	0.408	0.582	0.615
Constant	0.847	0.000	0.471

Table 9: Aligned dimension comparison. MAE/QWK use the same 295 dialogues with valid profiles from every model. All exact instead uses all 634 dialogues and seven ratings each, counting missing or out-of-range ratings as incorrect. The common subset is selected by model output validity and may be easier than the full test set.

the supplement.

7.3 Dimension Prediction and Output Validity

GPT, DeepSeek, and the supervised baselines return range-valid dimension profiles for all 634 dialogues after syntax normalization. Qwen does so for 295 dialogues (46.5%); 339 contain at least one rating outside 0–3. Those values are retained and counted as invalid, rather than clipped. Table 8 reports valid-profile metrics and coverage.

Table 9 compares all methods on the same 295 dialogues. GPT has the lowest MAE (0.393), while word TF–IDF has the highest mean QWK (0.656). This subset is selected by output validity and need not represent the full test set. To retain the full denominator, the table also reports exact rating accuracy over all 634×7 judgments, counting invalid ratings as incorrect. GPT obtains 0.667 and word TF–IDF 0.618; Qwen obtains 0.331 under this fixed prompting configuration. On the full test set, GPT and word TF–IDF have no clear difference in dimension MAE (paired 95% CI: $[-0.022, 0.036]$), while GPT has higher exact rating accuracy (difference 0.049; CI: $[0.026, 0.071]$).

Across tasks, no method leads every metric. The labels support distinctions between detecting pressure, naming its strategy, and estimating its degree. These cross-model comparisons do not isolate the benefit of training a given language model; the following diagnostic compares Qwen3-4B before and after training.

7.4 Matched Fine-Tuning Diagnostic

Table 10 compares the original Qwen3-4B with the checkpoint fine-tuned on the training split. The two runs use the same test inputs and decoding instruction. The reported valid n is the common number of dialogues with task-valid outputs; no missing or out-of-range values are repaired. "SFT" denotes the LoRA fine-tuned checkpoint.

Fine-tuning changes little across the four tasks. Pressure F1 is unchanged, intensity is nearly unchanged, and both strategy scores are slightly lower. Mean dimension QWK increases from 0.305 to 0.312 on the common valid subset, but the absolute gain is small. This result does not support a claim of broad performance improvement from fine-tuning; it instead shows that the fine-grained labels remain difficult even when the model is adapted to the training split.

8 Conclusion

LINGUAFORCE provides human judgments of pressure presence, intensity, strategies, and seven dimensions for 4,066 dialogues. Its labels support evaluation beyond binary detection, including whether models identify the kind and degree of pressure expressed in context. Three zero-shot language models and supervised lexical baselines perform strongly on binary detection, while strategy and dimension results reveal different strengths and output reliability. A matched Qwen3-4B fine-tuning run produces no broad gain across tasks, with only a small improvement in mean dimension QWK. This result supports using the resource to measure fine-grained understanding rather than assuming that adaptation makes every task easy. Independent annotation agreement remains necessary to assess the reliability of the human reference labels.

Limitations

The benchmark covers English dialogue and negative pressure. It does not measure listener compliance, psychological harm, or positive influence.

Task	Metric	Base	SFT	SFT–Base	Paired 95% CI
Pressure	F1	0.887	0.887	−0.0002	[−0.0030, 0.0024]
Intensity	MAE ↓	1.943	1.940	−0.0032	[−0.0126, 0.0063]
Intensity	QWK ↑	0.214	0.216	+0.0016	[−0.0036, 0.0073]
Strategies	Micro-F1 ↑	0.588	0.586	−0.0015	[−0.0040, 0.0010]
Strategies	Macro-F1 ↑	0.496	0.495	−0.0011	[−0.0042, 0.0024]
Dimensions	MAE ↓	0.878	0.874	−0.0039	[−0.0096, 0.0011]
Dimensions	Mean QWK ↑	0.305	0.312	+0.0069	[0.0009, 0.0135]

Table 10: Matched Qwen3-4B fine-tuning comparison on the held-out test set. The paired valid sample sizes are 634 for pressure and intensity, 623 for strategies, and 622 for dimensions. Intervals use 1,000 paired dialogue-level bootstrap resamples; they quantify uncertainty from the observed test sample for this fixed training run.

The strategy inventory and scales are operational choices whose reliability requires human agreement evidence.

Some labels have few positive examples, limiting the precision of per-label evaluation. The current duplicate check detects exact normalized matches but does not rule out paraphrases or shared scenario templates. The fine-tuning comparison uses one Qwen3-4B run and one fixed configuration; it does not estimate variation across random seeds, model sizes, or training recipes. Generalization to other languages, settings, and naturally occurring interactions requires additional evaluation. The comparison also does not cover other foundation-model families, such as Llama or GLM, whose behavior may differ under the same tasks (Touvron et al., 2023; GLM Team et al., 2024).

Ethical Considerations

The dialogues contain threats, shaming, deception, and verbal abuse. Predictions should not be used to diagnose individuals or infer intent or harm beyond the text. Annotators and data users should be informed about potentially distressing content. Participation was voluntary and unpaid. The dataset creators permit release of the dialogue texts and annotations. The anonymous supplement provides the data and experimental materials for review. The dialogues concern everyday scenarios and should not be treated as verified accounts of identifiable individuals. Language-model assistance was used for manuscript drafting and editing and for preparing experimental code. The benchmark reference labels were provided by human annotators.

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A Uncertainty Estimates

Paired comparisons. For strategy recognition, GPT minus word TF–IDF has a 95% interval of $[0.012, 0.062]$ for macro-F1 and $[-0.068, -0.037]$ for micro-F1. For dimension MAE, the corresponding interval includes zero on both the full test set and the shared valid-profile subset. These comparisons support task- and metric-specific conclusions rather than a single overall model ranking.

Interval estimation. All intervals are pointwise 95% intervals, without a multiple-comparison adjustment. Bootstrap intervals use the 2.5th and 97.5th percentiles of 1,000 dialogue resamples with replacement. Differences use paired draws, not subtraction of marginal interval endpoints. They are conditional on the observed split and predictions and do not establish stability across training seeds, API runs, or unseen settings.

For the API and lexical baselines, intensity and strategy intervals use all 634 test dialogues (Table 12). Dimension MAE and mean QWK are reported both on the shared 295-dialogue valid-profile subset (Table 13) and on all 634 dialogues for methods with complete valid profiles (Table 14). Exact rating accuracy always uses all 634 dialogues and counts invalid ratings as incorrect. Resampling whole dialogues preserves dependence among their seven ratings and fifteen strategy labels.

Mean dimension QWK is undefined if any of its seven component scores is undefined. Such

Model	FP	FPR (%)	95% CI
GPT-6 Astra	28	22.2	[15.8, 30.2]
Qwen3.8-Max	35	27.8	[20.7, 36.2]
DeepSeek-Flash	43	34.1	[26.4, 42.8]
Word TF–IDF	38	30.2	[22.8, 38.7]
Character TF–IDF	33	26.2	[19.3, 34.5]
Constant	126	100.0	[97.0, 100.0]

Table 11: False positives on 126 test dialogues labeled no pressure and request/advice or rational persuasion. Each dialogue is counted once; all systems return valid binary decisions. Intervals are Wilson intervals.

draws are excluded from that interval and counted in the tables. Paired QWK differences retain only draws defined for both methods. The shared subset has sparse dimension categories, so these conditional QWK intervals should not be interpreted as a stable model ranking. No undefined value is replaced with zero. All MAE, strategy-F1, and exact accuracy draws are defined. The supplementary JSON includes paired differences against both lexical baselines and counts of defined draws.

Non-pressuring influence. Among the 130 human-labeled no-pressure test dialogues, 125 carry a request/advice label and 45 a rational-persuasion label. Their union contains 126 dialogues, selected independently of model predictions. Table 11 reports false-positive rates with Wilson 95% intervals. The strategy labels are not independent annotations of action requests, so this diagnostic does not rule out lexical or scenario shortcuts. Supplementary results provide each group, all 130 negatives, and false-positive identifiers. No model is refitted for this analysis.

Model	Intensity MAE	Intensity QWK	Strategy macro-F1	Strategy micro-F1
GPT-6 Astra	[0.550, 0.672]	[0.752, 0.827]	[0.592, 0.626]	[0.672, 0.699]
Qwen3.8-Max	[0.823, 0.956]	[0.672, 0.753]	[0.572, 0.606]	[0.652, 0.681]
DeepSeek-Flash	[0.923, 1.068]	[0.580, 0.678]	[0.578, 0.612]	[0.662, 0.692]
Word TF–IDF	[0.631, 0.740]	[0.702, 0.768]	[0.550, 0.594]	[0.722, 0.751]
Character TF–IDF	[0.593, 0.696]	[0.732, 0.794]	[0.546, 0.591]	[0.716, 0.747]
Constant	[1.024, 1.200]	[0.000, 0.000]	[0.044, 0.049]	[0.217, 0.245]

Table 12: Pointwise 95% bootstrap intervals on all 634 test dialogues, with 1,000 paired dialogue resamples. All draws are defined. Point estimates appear in Tables 6 and 7.

Model	MAE (295)	QWK (295)	Defined QWK draws	Exact accuracy (634)
GPT-6 Astra	[0.352, 0.437]	[0.546, 0.635]	942	[0.651, 0.683]
Qwen3.8-Max	[0.578, 0.686]	[0.487, 0.586]	1000	[0.311, 0.354]
DeepSeek-Flash	[0.539, 0.656]	[0.465, 0.572]	1000	[0.486, 0.527]
Word TF–IDF	[0.377, 0.446]	[0.548, 0.691]	876	[0.600, 0.635]
Character TF–IDF	[0.376, 0.440]	[0.542, 0.615]	628	[0.596, 0.632]
Constant	[0.798, 0.897]	[0.000, 0.000]	628	[0.453, 0.489]

Table 13: Dimension 95% intervals. MAE/QWK use the same 295 complete-profile dialogues; exact rating accuracy uses all 634 dialogues, counting invalid outputs as wrong. QWK intervals exclude draws in which any dimension has undefined QWK; the resulting conditional intervals require caution. MAE and accuracy use all 1,000 draws.

Model	MAE	QWK	Defined QWK draws
GPT-6 Astra	[0.402, 0.459]	[0.571, 0.640]	1000
DeepSeek-Flash	[0.600, 0.671]	[0.515, 0.578]	1000
Word TF–IDF	[0.399, 0.445]	[0.584, 0.675]	1000
Character TF–IDF	[0.403, 0.445]	[0.580, 0.661]	999
Constant	[0.726, 0.789]	[0.000, 0.000]	993

Table 14: Dimension 95% intervals for methods with complete valid profiles on all 634 dialogues. Qwen is excluded from this full-profile comparison. Undefined QWK draws are excluded and counted; MAE uses all 1,000 draws.